

## Business Context

DeltaSquare is an NGO that works with the government on matters of social policy to bring about a change in the lives of underprivileged sections of society. They are given a task of coming up with a policy framework using a dataset that the government received from WHO.

## Objective

You, as a Data Scientist at DeltaSquare, are tasked with analyzing the data provided to identify the different factors that influence the income of an individual, build a prediction model that can help the government to formulate policies for the right pockets of the society, and share a proposal for the government.

## Data Dictionary

The data contains characteristics of the people as given below

1. age: continuous - age of a Person
2. workclass: Where do a person works - categorical -Private, Self-emp-not-inc, Self-emp-inc, Federal-gov, Local-gov, State-gov, Without-pay, Never-worked.
3. fnlwgt: This weight is assigned by the Current Population Survey (CPS). People with similar demographic characteristics should have similar weights since it is a feature aimed to allocate similar weights to people with similar demographic characteristics - continuous
4. education: Degree the person has - Bachelors, Some-college, 11th, HS-grad, Prof-school, Assoc-acdm, Assoc-voc, 9th, 7th-8th, 12th, Masters, 1st-4th, 10th, Doctorate, 5th-6th, Preschool.
5. education-num: no. of years a person studied - continuous.
6. marital-status: Married-civ-spouse, Divorced, Never-married, Separated, Widowed, Married-spouse-absent, Married-AF-spouse.
7. occupation: Tech-support, Craft-repair, Other-service, Sales, Exec-managerial, Prof-specialty, Handlers-cleaners, Machine-op-inspct, Adm-clerical, Farming-fishing, Transport-moving, Priv-house-serv, Protective-serv, Armed-Forces.
8. race: White, Asian-Pac-Islander, Amer-Indian-Eskimo, Other, Black.
9. sex: Female, Male.
10. capital-gain: Investment gain of the person other than salary - continuous
11. capital-loss: Loss from investments - continuous
12. hours-per-week: No. of hours a person works - continuous.

13. native-country: United-States, Cambodia, England, Puerto-Rico, Canada, Germany, Outlying-US(Guam-USVI-etc), India, Japan, Greece, South, China, Cuba, Iran, Honduras, Philippines, Italy, Poland, Jamaica, Vietnam, Mexico, Portugal, Ireland, France, Dominican-Republic, Laos, Ecuador, Taiwan, Haiti, Columbia, Hungary, Guatemala, Nicaragua, Scotland, Thailand, Yugoslavia, El-Salvador, Trinidad&Tobago, Peru, Hong, Holand-Netherlands.
14. salary: >50K, <=50K (dependent variable, the salary is in Dollars per year)